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AI-WEEK

Vibe Coding Hackathon

Team Details:

Team Name: *DATUM DECODERS*

Team Member Name and Phone Number:

Sai Sohan: 6309527147

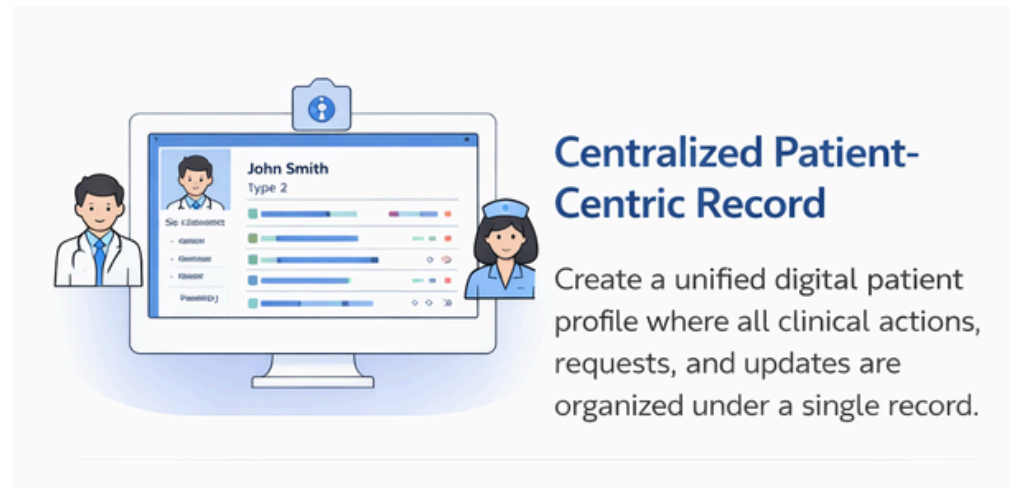
Sai Smaran: 9398446301

Suhas Belgavi: 9032496699

Problem Statement

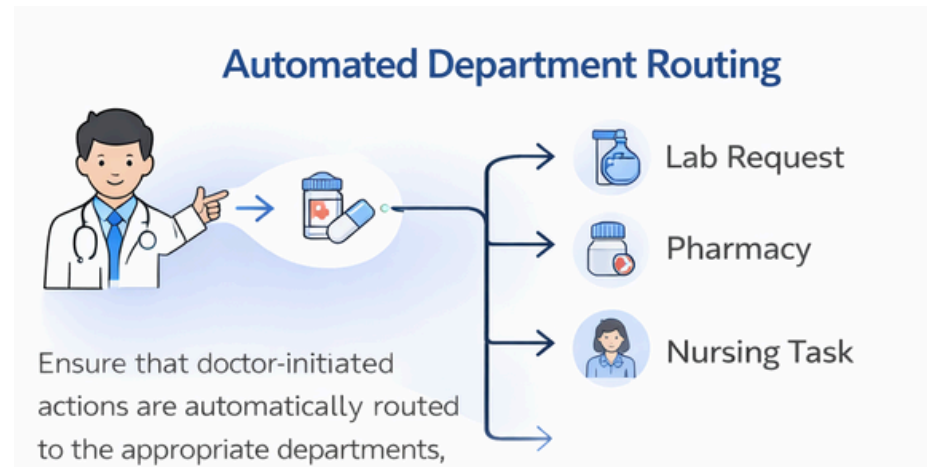
- Fragmented hospital workflows lead to manual handling of requests, delayed updates, and scattered patient information, making coordination difficult.
- Single patient-centered record to organize all clinical actions and requests, providing a shared view of each patient's care status.
- Doctor-initiated actions such as prescriptions, diagnostic requests, referrals, and care instructions are routed to appropriate departments and added to the patient care timeline.
- Real-time updates from nurses, diagnostic units, and pharmacies are immediately visible to all stakeholders involved in the patient's treatment.
- Live workflow visibility shows what actions are completed, in progress, and pending, ensuring smoother coordination and reduced delays.
- Expected outcomes: patient-wise records with real-time workflow tracking, clear departmental action tracking, and an end-to-end demo of coordination (doctor → diagnostics/pharmacy → update).

Proposed Solution



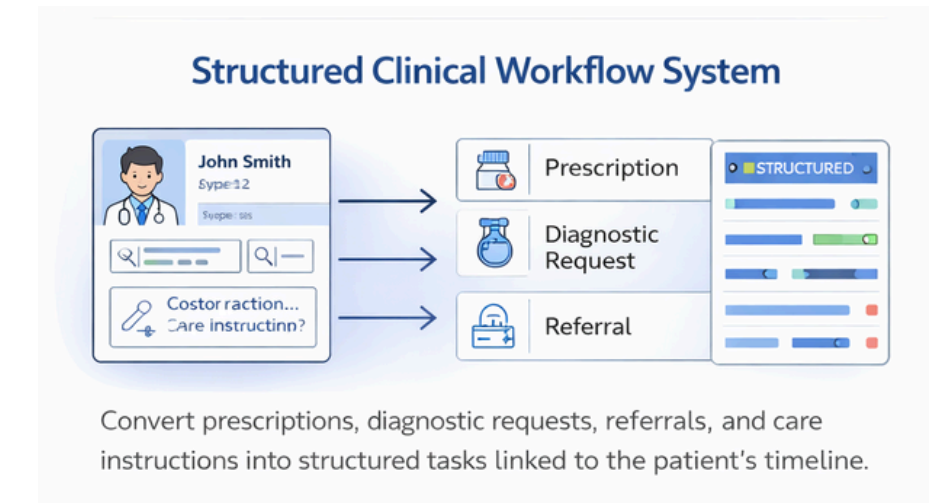
Centralized Patient Centric Record

- Establish a unified digital profile for every patient
- Consolidate all clinical actions within one centralized record
- Provide real-time updates and comprehensive care visibility



Automated Department Routing

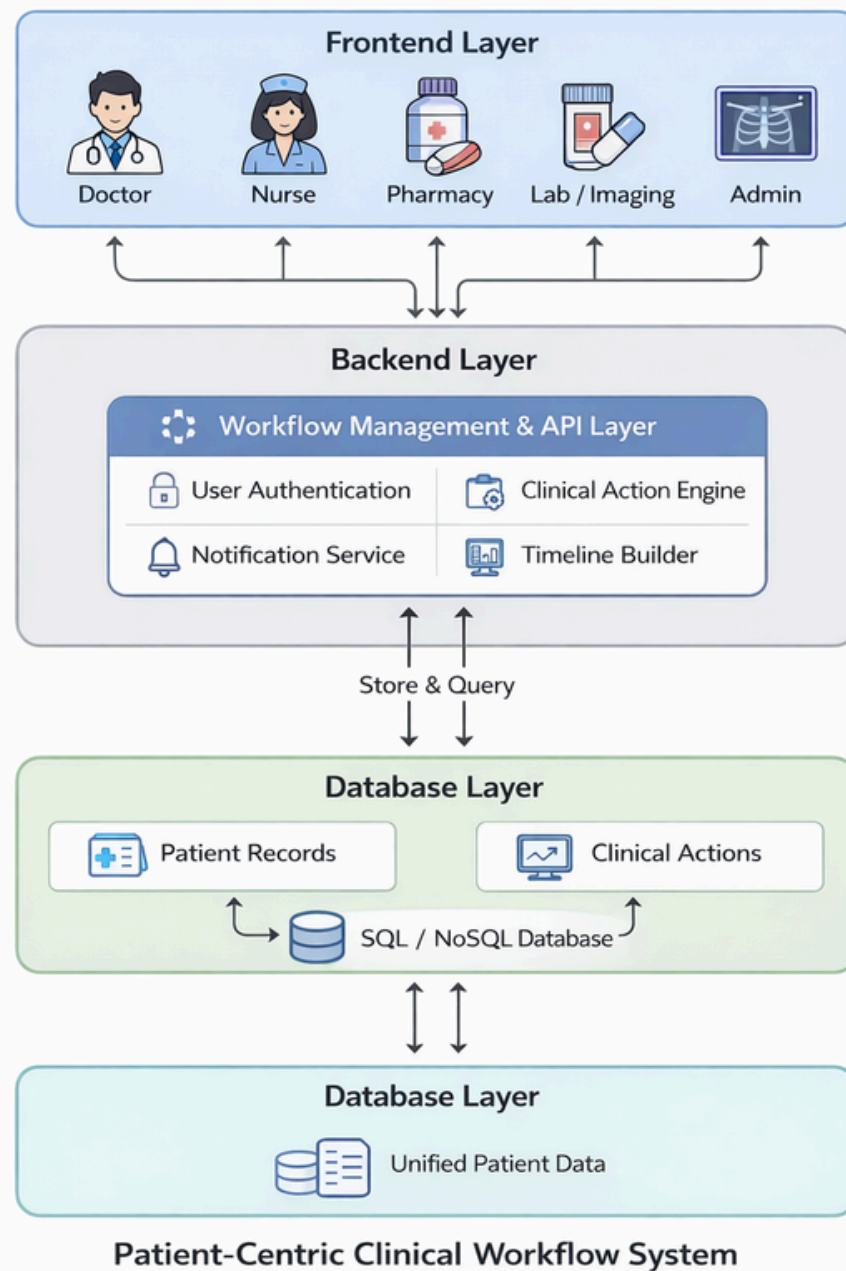
- Automatically route doctor-initiated requests to relevant departments
- Deliver lab, pharmacy, and nursing tasks without manual intervention
- Enable faster execution through intelligent department-based workflow routing



Structured Clinical Workflow System

- Convert clinical instructions into structured, trackable digital tasks
- Link prescriptions, diagnostics, and referrals to patient timeline

System Architecture



Layer	Description
Frontend Layer	Doctors, nurses, labs, pharmacy, and admin log in through role-based dashboards to create and receive clinical actions.
Backend Layer	Acts as the system brain – manages authentication, routes clinical requests, updates patient timelines, and sends real-time
Database Layer	Stores unified patient data, clinical actions, and status updates – serving as the single source of truth.

Tech Stack

Layer / Module	Key Components	Tech / Function
Front-End & User Access	Doctor Dashboard, Nurse Panel, Lab Queue, Pharmacy Panel, Admin Console	React.js Tailwind CSS, Role-based Routing, Socket.IO (real-time UI updates)
Backend Workflow Engine	Authentication, Clinical Action Engine, Timeline Builder, Notification Service	Node.js + Express / FastAPI, JWT Auth, REST APIs, Socket.IO / WebSockets
Data Management	Patient Records DB, Clinical Actions DB, User DB, Logs	Supabase/ PostgreSQL, Mongoose / Prisma ORM
Real-Time Coordination	Department Routing, Status Updates, Alerts	Socket.IO
Deployment & Security	Hosting, API Protection, Encryption	Vercel / Render / Railway, HTTPS, JWT, bcrypt

Thank You